

NOTE

RECOVERY OF *ACROPORA PALMATA* IN CURAÇAO:
A COMPARISON WITH THE FLORIDA KEYS

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ABSTRACT

Demographic monitoring of fixed plots of *Acropora palmata* (Lamarck, 1816) in Curaçao and the Florida Keys before and after acute tropical storm disturbance allowed for a comparison of recovery processes. The Curaçao population was characterized by greater colony size, live tissue abundance, and genotypic richness than the Florida Keys population, and hence was hypothesized to display a faster rate of recovery. However, after each population lost approximately half of its live tissue as a result of disturbance, magnitude of recovery differed significantly between regions, but was inconsistent with this hypothesis. Despite high post-disturbance asexual recruitment of fragments in Curaçao [12.1 (SE 4.9) recruits/plot compared to 4.6 (SE 1.5) for Florida Keys], consistent with the original hypothesis, Curaçao populations showed negligible recovery in terms of live tissue abundance (–2%) and have declined in number of colonies, whereas the Florida Keys population showed a slight, but significant increase in live tissue (13%) over the surveyed recovery period (from 7 to 31 mo post-disturbance).

Caribbean acroporids were the dominant reef-building coral in shallow reef zones before the late 1970s when disease, coupled with hurricanes and local anthropogenic impacts, precipitated a region-wide decline in Caribbean corals (Gardner et al. 2003). Caribbean acroporids, in particular, declined by >90% in many localities (Reviewed in Bruckner 2002, *Acropora*-Biological-Review-Team 2005). Due to past and continuing population decline and the lack of observed recovery, *Acropora palmata* (Lamarck, 1816) and *Acropora cervicornis* (Lamarck, 1816) were listed in 2006 as threatened under the US Endangered Species Act (ESA; NMFS 2006), and recently have been proposed for reclassification as endangered (NMFS 2012). In addition, they are designated as critically endangered on the IUCN red list (Carpenter et al. 2008).

Because disturbances, such as damaging storms and disease, are expected to continue and potentially increase in frequency and intensity (Harvell et al. 2002, Trenberth et al. 2007), it is important to understand a population's potential for recovery from disturbance events. Population sustainability is determined not only by its resistance to disturbance, but also by its ability to recover to pre-disturbance condition. Recovery depends on the population's capacity to recruit and maintain an adequate growth rate that outpaces the return time of disturbance. Population characteristics such as colony size, density, and genetic structure are expected to affect the potential for population recovery. For *A. palmata*, colony size relates indirectly to both sexual and asexual recruitment. Large-sized colonies produce more gametes (Soong and Lang 1992) and have a greater number of branches, yielding a higher contribution of both sexual and asexual propagules

(fragments). High colony densities may aid in recovery by providing a greater number of colonies to contribute to available fragments, while a combination of colony density and size, and the resultant structural complexity, may aid in fragment retention (Lirman and Fong 1997, Baums et al. 2003, Williams et al. 2008). Genotypic richness plays an important role, as *A. palmata* is an obligate outcrosser (Baums et al. 2005), so multiple genets in close proximity are necessary for fertilization and viable larval production.

Furthermore, populations with high genotypic richness may prove more robust as differences among genotypes may serve as biological “insurance” (e.g., differing susceptibility or capacity for successful recruitment) in the face of environmental disturbance. In Caribbean *Acropora* spp., particular genotypes have been shown to be resistant to disease (Vollmer and Kline 2008) or thermally-induced bleaching (Rogers and Muller 2012). Different coral genotypes display dissimilar responses in the production of non-calcifying tissue, which may affect fragment attachment rates (Frank et al. 1995). Such variation in individual genotypes may increase a population’s ability to resist perturbations and aid in recovery. Observations in other clonal organisms further suggest that high genotypic diversity can confer resistance to disturbance and/or greater recovery potential (Reusch et al. 2005, Hughes and Stachowicz 2009).

Geographic variation in population characteristics for *A. palmata* has been documented throughout the Caribbean region. Genetically distinct *A. palmata* populations in the eastern Caribbean have higher colony abundance (i.e., density), genotypic richness, and contribution via sexual reproduction compared with those in the western Caribbean (Baums et al. 2006). These characteristics suggest that eastern Caribbean populations may be more resistant to disturbance and resilient in recovering from disturbance than those in the western Caribbean. Populations in Curaçao, despite having suffered major declines (Nagelkerken and Nagelkerken 2004), are presumed to be typical of the eastern Caribbean, whereas those in the Florida Keys show lower genotypic richness, colony density, live tissue abundance, and smaller colony size, typical of the western province (Baums et al. 2006).

In the present study, we use demographic monitoring data from Curaçao, Netherlands Antilles, and the Florida Keys, USA, to (1) validate the ecological and genotypic differences previously documented between *A. palmata* populations from our two regional study populations, (2) quantify the effects of an acute disturbance event in each region, and (3) evaluate the hypothesis that an *A. palmata* population with greater genotypic richness, tissue abundance, and numbers of large colonies (Curaçao) will display a more rapid recovery than a more depauperate population (Florida Keys) following a comparable disturbance.

MATERIALS AND METHODS

Following a replicable protocol (Williams et al. 2006), a demographic monitoring study of Florida Keys *A. palmata* was initiated in 2004 to quantify sources and rates of both mortality and recruitment. This protocol depends on identifying and tracking all individual colonies within a plot and thus, necessarily targets stands with an intermediate colony density (fewer than 50 colonies per plot) at the time of setup. A comparable study of the *A. palmata* population in Curaçao was implemented in 2006 (Kramer et al. 2009). Populations studied in both regions were located in the typical *A. palmata* habitat on the

reef crest or fore-reef in depths shallower than 7 m. The *A. palmata* population in the Florida Keys occupies fore-reef spur-and-groove formations approximately 10 km from the coast on a barrier reef, whereas the populations in Curaçao occupy fringing reef habitat. Both populations are directly exposed to open-ocean wave action.

In both regions, fixed 150 m² study plots were established at haphazardly selected intermediate-density *A. palmata* stands (mean 23; range 6–53 colonies per plot) with three plots on each of three reefs along the southern coast of Curaçao ($n = 9$) and 15 plots among five reefs in the upper Florida Keys (Table 1). Plots were monitored annually to document trends in abundance, recruitment, the prevalence of disease, and the corallivorous snail, *Coralliophila abbreviata* (Lamarck, 1816) (referred to as “snails” hereafter). Both locations experienced a major storm disturbance shortly after monitoring began, allowing us to contrast population responses. The Florida Keys population was affected by hurricanes Dennis, Katrina, Rita, and Wilma during the months of July through October 2005 (Williams et al. 2008), and the Curaçao populations by hurricane Omar in October 2008. The impact of individual hurricanes is patchy and inconsistent, and storm intensity and frequency does not necessarily parallel degree of reef damage (Harmelin-Vivien 1994). Therefore, we chose to characterize disturbance by the magnitude of damage inflicted on *A. palmata* rather than by storm characteristics, and the multiple hurricanes in Florida Keys were treated as a single overall disturbance event.

Three annual post-disturbance surveys were conducted each spring in each region (2006–2008 in Florida and 2009–2011 in Curaçao). Surveys 1, 2, and 3 were conducted 7/8 (Florida/Curaçao), 19, and 31 mo post-disturbance. Some results of the demographic monitoring in the Florida population have been reported previously (Williams et al. 2008, Williams and Miller 2012). It should be noted that although reefs of the Florida Keys were affected by a moderate bleaching event in 2005 (Wagner et al. 2010), *A. palmata* in our study population were not affected (Williams and Miller 2012).

All *A. palmata* colonies in each study plot (delineated according to colony definitions established in Williams et al. 2006) were mapped from a central stake, photographed (to facilitate identifying the colony over time), and measured (length, width, and height). In addition, a single observer over the course of the study made a visual estimate of percent live tissue cover of each colony. A subset of approximately 12 colonies in each plot was randomly selected (all colonies were selected if fewer than 15 were present at the start of monitoring) and tagged. Observations of various conditions/threats (e.g., recent mortality, disease, presence of corallivorous snails) and snail abundance (number of snails found on a colony) were recorded for tagged colonies. Disease prevalence and the prevalence of snail-occupied colonies were based on the subset of tagged colonies within each plot.

During annual surveys, photographs and coordinates of colonies mapped during the previous survey were used to identify those colonies that were still present. New colonies (typically <10 cm in diameter or resembling a branch fragment) were mapped as “recruits” within each plot, and those found dead or missing were recorded accordingly. In some instances (e.g., extreme fragmentation events in plots BB1 and SQ2 in 2007; SQ2 in 2009), re-identifying all colonies was not possible and the plots were re-mapped, thus only a net change in number of colonies could be calculated for the previous survey interval for each plot. Sexual recruitment was not discernible in the present study, and therefore, hereafter, the terms recruit and recruitment refer to any new attached colony including asexual recruitment resulting from a live fragment attaching to the substrate.

Baums et al. (2006) contrasted the genotypic structure in populations of *A. palmata* in Curaçao and the Florida Keys. These results were ground-truthed for our specific study plots by documenting genotypic richness, defined here as the number of distinct genets standardized to the number of colonies sampled (N_g/N). Small biopsy samples were collected from all tagged colonies in Curaçao and the Florida Keys in 2006 for multi-locus genotyping. Samples were analyzed at Penn State University utilizing the methods and microsatellite markers of Baums et al. (2005).

The dynamics of two aspects of *A. palmata* "abundance" (live tissue and number of colonies) and mean colony size were examined to characterize initial population status and compare the changes associated with disturbance and recovery between the two regions. The mean of the colony length, width, and height ("colony average dimension") was used as an index of colony size. Live tissue abundance was estimated for each colony as a "live area index" (LAI) by squaring its average dimension and multiplying this area by the visual estimate for "percent live" for that colony. LAI was then summed for all colonies in the plot (plot LAI) to estimate tissue abundance for the plot (see Williams et al. 2008).

T-tests (when assumptions were validated) or Mann-Whitney U-tests (when not) were used for statistical comparisons between regions of initial population characteristics and the change in population characteristics following the disturbance event using plots as replicates. Four initial population characteristics (genotypic richness, tissue abundance, colony density, and mean colony size) were compared under objective one (documenting ecological/genotypic differences in the initial condition of the two populations). The percent change in abundance and size parameters between the survey prior and the survey following the disturbance in each region were compared to examine the relative magnitude of the disturbance effects between the regions, despite regional differences in the timing, frequency, and storm strength. The overall prevalence (percent of colonies surveyed, averaged over all surveys) of disease and snail occupation did not conform to the assumptions of parametric testing, so Mann-Whitney U-tests were used to compare between regions.

To test our hypothesis that Curaçao populations recovered more quickly than Florida Keys populations, we used a two-way repeated-measures ANOVA when assumptions were validated (plot LAI), or separate one-way nonparametric repeated-measures ANOVAs (Friedman tests) for each region when assumptions were not validated (colony density and mean colony size). For the two-way repeated-measures ANOVA, the factors were region (Curaçao versus Florida Keys) and time (post-disturbance survey, three levels). Where significant factors were identified, post-hoc (Holm-Sidak) tests were performed to determine which treatments were significantly different ($P < 0.05$). Note that n was reduced from nine to eight for this recovery phase in Curaçao as one plot, SM3, was extirpated by the disturbance.

RESULTS AND DISCUSSION

Plots of population characteristics with respect to time of the disturbance in each region indicated that mean colony densities were similar pre-disturbance (*t*-test: $P = 0.911$; Table 1). This was not unexpected as intermediate density stands were selected to facilitate colony tracking in the monitoring protocol. Populations in the two regions exhibited non-significant trends in mean live tissue (29% higher in Curaçao: mean of 9.7 vs 7.5 m² plot LAI in Florida; *t*-test: $P = 0.252$; Fig. 1A) and mean colony dimension (25% greater in Curaçao: 70.6 vs 56.4 cm; *t*-test, $P = 0.058$; Fig. 1C). However, genotypic richness was over three times greater in Curaçao (Mann-Whitney U-test: $P < 0.001$ for Ng/N ; Table 1). Thus, the regionally-formulated hypothesis based on improved ecological and genotypic status of populations in Curaçao vs Florida was largely appropriate within our study plots.

Following the hurricane disturbance, *A. palmata* populations in both regions experienced substantial losses of very similar magnitude (Curaçao plot LAI, -55%; Florida plot LAI, -56%; *t*-test: $P = 0.985$; Fig. 1A). Average colony dimension also decreased by 37% in Curaçao and 19% in Florida populations (not significantly different between the two regions; *t*-test: $P = 0.065$; Fig. 1C). Thus, although the details of hurricane exposure at the two sites differed in timing, number, and intensity, the magnitude of the disturbance event evidenced by the amount of

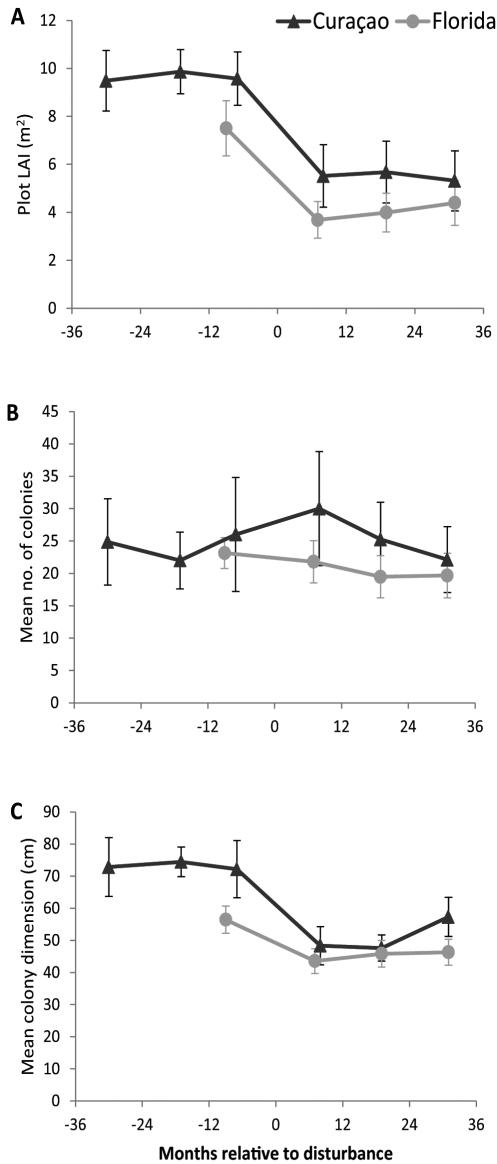


Figure 1. Temporally shifted data comparing *Acropora palmata* population characteristics relative to hurricane disturbance (month 0) in Curaçao ($n = 9$ permanent 150 m² plots; time 0 = fall 2008) and Florida ($n = 15$ plots; time 0 = fall 2005). (A) Average tissue abundance summed for all colonies in a plot (Plot LAI, see text for definition), (B) mean number of colonies per plot, and (C) mean colony dimension. Error bars represent one standard error.

live tissue lost was virtually identical, allowing for the comparison of population recovery from comparable-magnitude disturbance between the two populations with initially contrasting ecological and genotypic characteristics.

Despite these declines in tissue abundance and colony size, colony abundance increased somewhat in the first survey following the disturbance (i.e., 7–8 mo after) in Curaçao (+5%), whereas it declined slightly in Florida (–4%; Fig. 1B). Though

Table 1. Location, colony abundance [initial number of colonies (randomly tagged colonies) per 150 m² area study plot], and genotypic richness (number of unique genets standardized to the number of colonies sampled, Ng/N) sampled at the initiation of the study (2006 in Curaçao, 2004 in Florida). Only tagged colonies were sampled for multi-locus genotyping.

Site	Plot	Latitude	Longitude	Initial # colonies	Ng/N
CURAÇAO					
Blue Baaï	BB1	12.13511	-68.98682	13 (9)	0.78
	BB2	12.13527	-68.98710	16 (13)	0.54
	BB3	12.13520	-68.98738	10 (9)	0.89
Boca Santa Marta	SM1	12.26790	-69.12822	16 (16)	0.82
	SM2	12.26790	-69.12822	6 (6)	1.00
	SM3	12.26683	-69.12720	15 (14)	0.14
Sea Aquarium Reef	SQ1	12.08417	-68.89493	53 (12)	0.50
	SQ2	12.08365	-68.89467	41 (12)	0.58
	SQ3	12.08312	-68.89585	33 (12)	0.83
MEAN (SE)	—	—	—	23 (SE 5.3)	0.68 (SE 0.09)
FLORIDA					
Carysfort	CF1	25.22194	-80.21055	17 (13)	0.23
	CF2	25.22178	-80.21060	37 (12)	0.08
	CF3	25.22290	-80.20956	11 (10)	0.10
Elbow	EL1	25.14259	-80.25835	43 (11)	0.36
	EL2	25.14290	-80.25822	26 (12)	0.17
	EL3	25.14394	-80.25780	31 (12)	0.17
	EL4	25.14508	-80.25734	21 (13)	0.15
	EL5	25.14518	-80.25740	13 (10)	0.30
French	FR1	25.03393	-80.34959	29 (11)	0.36
Key Largo Dry Rocks	KL1	25.12360	-80.29736	17 (12)	0.08
	KL2	25.12290	-80.29787	27 (12)	0.25
	KL3	25.12255	-80.29826	15 (12)	0.25
Molasses	ML1	25.00958	-80.37481	14 (10)	0.50
	ML2	25.00912	-80.37473	23 (14)	0.14
	ML3	25.01015	-80.37328	23 (11)	0.18
MEAN (SE)	—	—	—	23 (SE 2.4)	0.22 (SE 0.03)

these differences were not significant (t -test: $P = 0.604$), the increase in Curaçao is likely explained by the initial recruitment of large numbers of new colonies [12 (SE 5) recruits/plot in Curaçao compared to 5 (SE 2) recruits/plot in Florida]. This greater initial recruitment, largely consisting of attached fragments, is consistent with the hypothesis of greater population resilience in Curaçao.

However, over the following two post-disturbance surveys (between survey 1 and 3; a total time span of 23 mo for Curaçao and 24 mo for Florida), recovery was slow to negligible in both regions. Two-way repeated-measures ANOVA on tissue abundance (i.e., plot LAI) indicated a significant interaction of region and recovery time ($P = 0.024$). Thus, the trajectory of tissue abundance over time (or recovery) varied between regions. Curaçao populations did not change significantly in tissue abundance from survey 1 to survey 3 [plot LAI -2% (SE 8%); pairwise Holm-Sidak post-hoc tests among surveys: $P > 0.05$; Fig. 1C], whereas the Florida population displayed significant but slow recovery in terms of tissue abundance

[plot LAI increased by 13% (SE 10%); post-hoc Holm-Sidak comparison for survey 1 to survey 3: $P = 0.001$]

For colony abundance, the Florida population showed a non-significant (Friedman test: $P = 0.062$) negative trend [-15% (SE 7%); Fig. 1B] and no significant change in colony size (Friedman test: $P = 0.165$; Fig. 1C). Meanwhile, Curaçao colony abundance showed a similar non-significant decreasing trend [-11% (SE 9%); Friedman test: $P = 0.227$; Fig. 1B], while colony size in Curaçao increased significantly over the recovery period from survey 1 to survey 3 [21% (SE 4%); Friedman test: $P = 0.008$; Fig. 1C]. Overall, results were not consistent with our hypothesis predicting greater recovery in Curaçao.

The paradox of Curaçao's positive initial recruitment and subsequent apparent increase in mean colony size with no increase in tissue abundance is likely explained by a "reversal of fortune." Size-frequency data show a loss of small-sized colonies (approximately half were lost between surveys 1 and 3), and negligible increases in frequencies of mid-sized colonies over this 24-mo recovery period (Appendix 1). Thus, we interpret that there was good initial recruitment of disturbance-generated fragments in Curaçao, consistent with our hypothesis, but the majority of these recruits subsequently died and did not contribute to population recovery. This observation highlights that for complex clonal species, a single population characteristic may show apparently positive trends but when viewed in light of other characteristics may reflect negative population processes, and thus could be misinterpreted when considered alone. For example, the increase in mean colony size observed in Curaçao could be a positive attribute for a recovering population, but this increase likely resulted from the loss of small-sized colonies rather than colony growth, which is not indicative of a recovering population. To provide an accurate depiction of recovery processes for *A. palmata*, multiple population characteristics should be considered in conjunction, such as colony abundance, live tissue abundance, and mean colony size.

In our study, disease and corallivory were examined as mechanisms known to impair *Acropora* spp. population recovery (e.g., Knowlton et al. 1990). Seasonally concurrent (i.e., spring) disease prevalence ("white disease" conditions pooled) was approximately three times higher in Curaçao (Mann-Whitney U-test: $P < 0.001$; Fig. 2), suggesting that disease may contribute to impaired recovery in Curaçao. In contrast, overall average snail prevalence (expressed as proportion of colonies occupied by snails) was approximately three times higher in Florida (Mann-Whitney U-test: $P < 0.001$; Fig. 2), suggesting a greater toll of predation on the Florida population. Snail abundance on infested colonies was similarly higher in Florida (approximately 2 times; data not shown). This contrasting pattern in disease vs predation prevalence between regions suggests that disease is more important than predation in limiting population expansion in Curaçao. It should be noted, however, that this regional comparison was made only in spring and *Acropora* diseases are known to display marked seasonality. Florida disease prevalence is actually at its annual low in spring (Williams and Miller 2012).

The monitoring protocol used here targets areas with intermediate densities of coral colonies (Williams et al. 2006), which begs the question whether the dynamics displayed in these plots are representative of the broader populations in each region. Our extensive observations in the upper Florida Keys and more limited observations along the south coast of Curaçao suggest that these stands are not

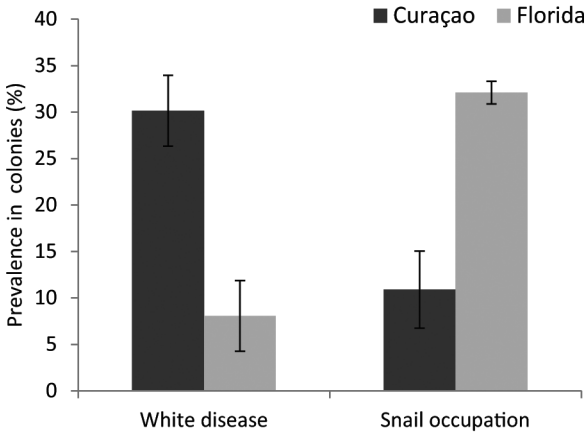


Figure 2. Disease and predation on *Acropora palmata*. Prevalence [mean (SE)] of “white disease” and occupation by corallivorous snails, *Coralliophila abbreviata*, in Curaçao and Florida averaged over plots and surveys ($n = 5$ surveys for each region).

atypical. Higher density stands may be present along the north coast of Curaçao (e.g., Baums et al. 2006), but to our knowledge they have not been well characterized in recent times.

Coral population recovery from disturbance proceeds over a longer time frame than the 2–3 yr recovery interval reported here. Williams and Miller (2012) estimated an 11-yr recovery time for the Florida population (to reach 2004 baseline of LAI), whereas the predicted return time of hurricane disturbance in the upper Florida Keys is 7 yrs (Blake et al. 2011). Although Curaçao’s expected frequency of hurricane disturbance is about 10 times less than the Florida Keys (Lugo et al. 2000), the lack of recovery over 2–3 yrs following the disturbance suggests an overall declining baseline, especially in the face of potentially increasing frequency and intensity of other acute mortality drivers, such as thermal stress (e.g., McWilliams et al. 2005) and disease (Harvell et al. 2002). A declining baseline for this foundation species has continued dire implications for ecological function and ecosystem services of coral reefs. High disease occurrence and lack of effective population recovery in the Curaçao population, which initially had characteristics predicted to confer higher resilience, underscores the acute need to develop research and management strategies to enhance fragment survivorship and mitigate disease effects.

ACKNOWLEDGMENTS

Funding was provided by the NOAA Coral Reef Conservation Program, the NMFS Southeast Regional Office, Royal Caribbean International, and the Education and Research Department of the Curaçao Sea Aquarium. Many thanks for field support in Curaçao from A Anton, L Johnston, A Valdivia, R Wilborn, and S Piontek.

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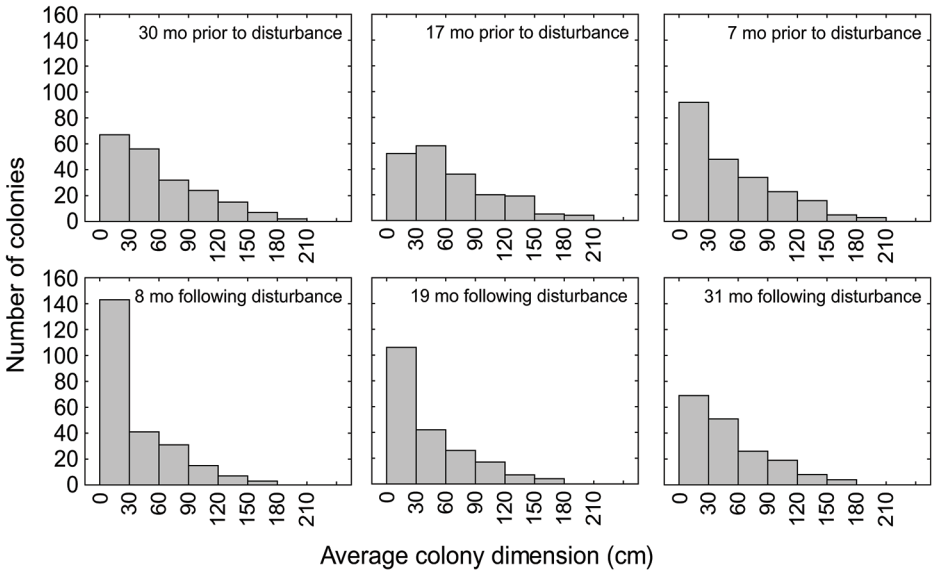
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DATE SUBMITTED: 24 April, 2012.

DATE ACCEPTED: 29 March, 2013.

AVAILABLE ONLINE: 18 April, 2013.

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Appendix 1. Size frequency histograms of all colonies in Curaçao at each survey based on maximum dimension of the colony (cm). Dramatic decline in small colonies has been observed since the hurricane disturbance without consequent increase in larger size classes. Surveys 1, 2, and 3 in the text refer to months 8, 19, and 31 following the disturbance.

